Difference in Difference

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A brief information about me

- Name: Jahen F. Rezki
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  - PhD in Economics (University of York, UK),
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- Position:
  - Assistant Professor at FEB UI;
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Course Logistics

- I'm going to teach about Difference in Difference
- Reference
  - Angrist, J. D., Pischke, J. S. (2009). Mostly harmless econometrics: An empiricist's companion. Princeton university press.
  - Cunningham, S. (2021). Causal inference: The mixtape. Yale university press.
- Important Source: Asjad Naqvi website: `https://asjadnaqvi.github.io/DiD/`

Course Logistics

- Please ask questions
- Please let me know if I'm talking too fast
- Please let me know if something is unclear
- I may not be able to answer all of your questions

Outline

- Differences in Differences: Basics
- Differences-in-Differences: Card & Krueger (1994)
- Regression Differences-in-Differences
- Combining Differences-in-Differences in a Panel Data Framework

False counterfactuals

Before vs. After Comparisons:

- Compares: same individuals/communities before and after the program
- Drawback: does not control for time trends

Participant vs. Non-Participant Comparisons:

- Compares: participants to those not in the program
- Drawback: selection — why didn't non-participants participate?

Two Wrongs Sometimes Make a Right

Difference-in-differences (or "diff-in-diff" or "DD") estimation combines the (flawed) pre vs. post and participant vs. non-participant approaches

- This can sometimes overcome the twin problems of [1] selection bias (on fixed traits) and [2] time trends in the outcome of interest

- The basic idea is to observe the (self-selected) treatment group and a (self-selected) comparison group before and after the program

The diff-in-diff estimator is:

$$DD = \bar{Y}^{treatment}_{post} - \bar{Y}^{treatment}_{pre} - (\bar{Y}^{comparison}_{post} - \bar{Y}^{comparison}_{pre})$$

DD estimation: early examples

1849: London's worst cholera epidemic claims 14,137 lives

- Two companies supplied water to much of London: the Lambeth Waterworks Co. and the Southwark and Vauxhall Water Co.
  - Both got their water from the Thames
- John Snow believed cholera was spread by contaminated water

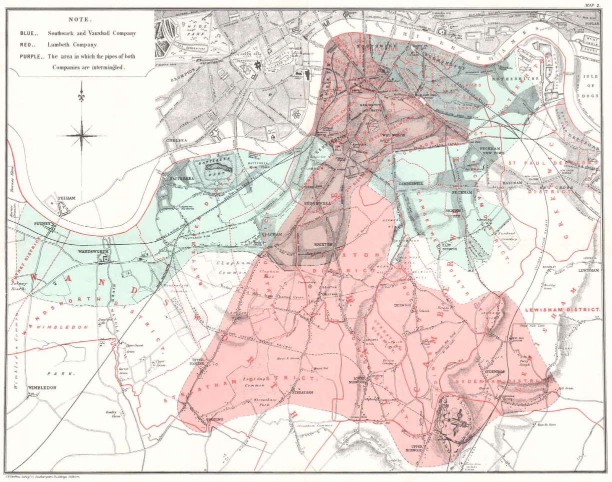
1852: Lambeth Waterworks moved their intake upriver

- Everyone knew that the Thames was dirty below central London

1853: London has another cholera outbreak

- Are Lambeth Waterworks customers less likely to get sick?

DD estimation: early examples



Source: John Snow Archive and Research Companion

DD estimation: early examples

John Snow's Grand Experiment:

- Mortality data showed that very few cholera deaths were reported in areas of London that were only supplied by the Lambeth Waterworks
- Snow hired John Whiting to visit the homes of the deceased to determine which company (if any) supplied their drinking water
- Using Whiting's data, Snow calculated the death rate
  - Southwark and Vauxhall: 71 cholera deaths/10,000 homes
  - Lambeth: 5 cholera deaths/10,000 homes
- Southwark and Vauxhall are responsible for 286 of 334 deaths
  - Southwark and Vauxhall moved their intake upriver in 1855

DD estimation: early examples

In the 1840s, observers of Vienna's maternity hospital noted that death rates from postpartum infections were higher in one wing than the other

- Division 1 patients were attended by doctors and trainee doctors
- Division 2 patients were attended by midwives and trainee midwives

Ignaz Semmelweis noted that the difference emerged in 1841, when the hospital moved to an "anatomical" training program involving cadavers

- Doctors received new training; midwives never handled cadavers
- Did the transference of "cadaveric particles" explain the death rate?

Semmelweis proposed an intervention: hand-washing with chlorine

- Policy implemented in May of 1847

DD estimation: early examples

	Physicians' Division			Midwives' Division		
		Deaths			Deaths	
Year	Births	No.	%	Births	No.	%
1841	3036	237	7.7	2442	86	3.5
1842	3287	518	15.8	2659	202	7.5
1843	3060	274	8.9	2739	169	6.2
1844	3157	260	8.2	2956	68	2.3
1845	3492	241	6.8	3241	66	2.03
1846	4010	459	11.4	3754	105	2.7
1847				3306	32	0.9
January–May	2134	120	5.6			
	Intervention introduced in May					
June–December	1841	56	3.04			
1848	3556	45	1.27	3219	43	1.33

DD estimation: early examples

BULLETIN OF THE

U. S. BUREAU OF LABOR STATISTICS.

WHOLE NO. 176. WASHINGTON. JULY, 1915.

EFFECT OF MINIMUM-WAGE DETERMINATIONS IN OREGON.[1]

BY MARIE L. OBENAUER AND BERTHA VON DER NIENBURG.

Source: Obenauer and Nienburg (1915)

DD estimation: early examples

In 1913, Oregon increased the minimum wage for experienced women to $9.25 per week, with a maximum of 50 hours of work per week

- Minimum wage for inexperienced women (and girls) also increased, but was new minimum ($6/week) not seen as a binding constraint
- Obenauer and Nienburg obtain HR records of 40 firms
- Compare the employment of experienced women before and after the minimum wage to law to employment of girls, inexperienced women, and men

DD estimation: early examples

TABLE 1.—ESTABLISHMENTS COVERED IN THE INVESTIGATION AND WOMEN AND MEN EMPLOYED DURING PERIOD STUDIED IN 1914.

[This table does not include extra male or female help whose identity from week to week could not be traced, such female help being equivalent to 3 women working full time; nor does it include 20 saleswomen whose regular employment began with the opening of a new department on the last day of the period covered in the investigation.]

Type of store.	Number of establishments covered.	Number of persons employed during period studied in 1914.	
		Women and girls.	Men.
PORTLAND.			
Department, dry-goods, and 5 and 10 cent stores	6	1,345	802
Specialty stores	11	181	49
Neighborhood stores	16	20	17
Total	33	1,546	868
SALEM.			
Dry-goods, specialty, and 5 and 10 cent stores	7	96	34
Grand total	40	1,642	902

[1] See note [1], p. 57.
[2] One firm, Olds, Wortman & King, a Portland department store, refused the Federal agents access to their records. They offered to furnish a summary statement, but the Bureau did not regard this as comparable with material obtained direct from other firms' books.

Source: Obenauer and Nienburg (1915)

DD estimation: early examples

TABLE 1
EMPLOYMENT CHANGES IN PORTLAND RETAIL STORES, 1913–1914

	Men	Girls (Age 16–18)	Ratio (Girls/Men)	Women (Age > 18)	Ratio (Women/Men)	Women Age Unknown
Before (Mar/Ap 1913)	940	138	.1468	1543	1.641	152
After Mar/Ap 1914)	868	160	.1843	1327	1.529	59
Change	–72	22	.0375	–216	–0.113	–93
% Change	–7.7%	15.9%	23.6%	–14%	–6.3%	–61.2%

Source: Obenauer and Nienburg (1915, Table 3, pages 14–15).

Source: Kennan (1995)

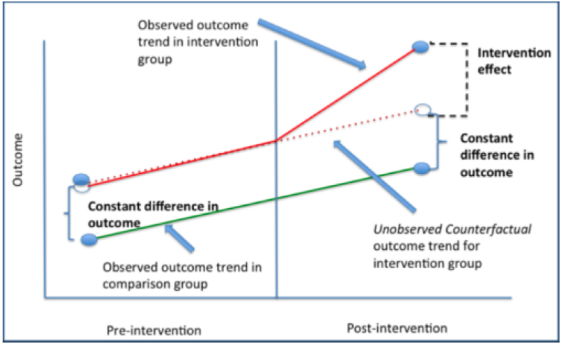
Difference-in-Difference estimation

	Treatment	**Comparison**
Pre-Program	$\bar{Y}^{treatment}_{pre}$	$\bar{Y}^{comparison}_{pre}$
Post-Program	$\bar{Y}^{treatment}_{post}$	$\bar{Y}^{comparison}_{post}$

Intuitively, diff-in-diff estimation is just a comparison of 4 cell-level means

- Only one cell is treated: Treatment $\times$ Post-Program

Basic Set up-Illustration



Three Ingredients

- A dummy for the treatment observation (T). It varies across observation and it controls for fixed differences between the units being compared.
- A dummy for post-treatment period (t). It varies over time and it controls for the fact that conditions change over time for all observations.
- The interaction term T*t. It is generated by multiplying the two dummies, the coefficient on this term is the DiD causal effect.

Card & Krueger (1994)

- Suppose you are interested in the effect of minimum wages on employment (a classic and controversial question in labour economics).
- In a competitive labour market, increases in the minimum wage would move us up a downward-sloping labour demand curve.
  - employment would fall.

Card & Krueger (1994)

- Card & Krueger (1994) analyse the effect of a minimum wage increase in New Jersey using a differences-in-differences methodology.

- In February 1992 NJ increased the state minimum wage from \$4.25 to \$5.05. Pennsylvania's minimum wage stayed at \$4.25.

- They surveyed about 400 fast food stores both in NJ and in PA both before and after the minimum wage increase in NJ.

Diff-in-Diff Strategy

- DD is a version of fixed effects estimation. Formally:

$$Y_{1ist} : \text{employment at restaurant } i, \text{ state } s, \text{ time } t \text{ with a high } w^{min}$$
$$Y_{0ist} : \text{employment at restaurant } i, \text{ state } s, \text{ time } t \text{ with a low } w^{min}$$

- We then assume that:

$$E[Y_{0ist}|s, t] = \gamma_s + \lambda_t$$

- In the absence of a minimum wage change, employment is determined by the sum of a time-invariant state effect γ_s and a year effect λ_t that is common across states (common trends assumption).

Diff-in-Diff Strategy

- Let D_{st} be a dummy for high-minimum wage states and periods
- Assuming $E[Y_{1ist} - Y_{0ist}|s, t] = \delta$ is the treatment effect, observed employment can be written:

$$Y_{ist} = \gamma_s + \lambda_t + \delta D_{st} + \epsilon_{ist}$$

Diff-in-Diff Strategy

In New Jersey:

- Employment in February is:

$$E[Y_{ist}|s = NJ, t = Feb] = \gamma_{NJ} + \lambda_{Feb}$$

- Employment in November is:

$$E[Y_{ist}|s = NJ, t = Nov] = \gamma_{NJ} + \lambda_{Nov} + \delta$$

- The difference between February and November is:

$$E[Y_{ist}|s = NJ, t = Nov] - E[Y_{ist}|s = NJ, t = Feb] = \lambda_{Nov} - \lambda_{Feb} + \delta$$

Diff-in-Diff Strategy

In Pennsylvania:

- Employment in February is:

$$E[Y_{ist}|s = PA, t = Feb] = \gamma_{PA} + \lambda_{Feb}$$

- Employment in November is:

$$E[Y_{ist}|s = PA, t = Nov] = \gamma_{PA} + \lambda_{Nov}$$

- The difference between February and November is:

$$E[Y_{ist}|s = PA, t = Nov] - E[Y_{ist}|s = PA, t = Feb] = \lambda_{Nov} - \lambda_{Feb}$$

Diff-in-Diff Strategy

- The differences-in-differences strategy amounts to comparing the change in employment in NJ to the change in employment in PA.
- The population differences-in-differences are:

$$\begin{aligned} E[Y_{ist}|s = NJ, t = Nov] - E[Y_{ist}|s = NJ, t = Feb] \\ -E[Y_{ist}|s = PA, t = Nov] - E[Y_{ist}|s = PA, t = Feb] = \delta \end{aligned}$$

- This is estimated using the sample analog of the population means.

Diff-in-Diff Table

	Stores by state		
Variable	PA (i)	NJ (ii)	Difference, NJ − PA (iii)
1. FTE employment before, all available observations	23.33 (1.35)	20.44 (0.51)	−2.89 (1.44)
2. FTE employment after, all available observations	21.17 (0.94)	21.03 (0.52)	−0.14 (1.07)
3. Change in mean FTE employment	−2.16 (1.25)	0.59 (0.54)	2.76 (1.36)

Surprisingly, employment rose in NJ relative to PA after the minimum wage change.

Main Results of Card and Krueger (1994)

- Minimum wage increase did not lead to job losses in fast-food restaurants in New Jersey compared to those in eastern Pennsylvania.
- Employment actually increased in New Jersey relative to Pennsylvania after the minimum wage increase.
- Wages increased for low-wage workers in New Jersey relative to Pennsylvania.
- There was no evidence of significant price increases at fast-food restaurants in New Jersey.
- The study triggered a long academic debate that's still very active because it challenged the conventional wisdom that raising the minimum wage leads to job losses.

Critiques of Card and Krueger (1994)

- Small sample size: Only 410 fast-food restaurants in New Jersey and eastern Pennsylvania.
- Unreliable data: Payroll records not designed for research purposes, may contain errors.
- Lack of statistical significance: Differences in employment levels between states not statistically significant.
- Unrepresentative comparison group: Comparison group may not be a good representation of control group.
- Theoretical limitations: Study did not account for potential long-term effects of minimum wage increases.

Follow up Study: Card and Krueger (2000)

- Re-analysis of original 1994 study with updated data and improved methodology.
- Confirmed previous findings that minimum wage increases do not lead to job losses in the fast-food industry.
- Expanded analysis to include more states and industries, finding no evidence of job losses due to minimum wage increases.
- The study challenged the view that minimum wage increases lead to job losses across all industries and regions.

Regression DD

- We can estimate the differences-in-differences estimator in a regression framework.
- Advantages:
  - It is easy to calculate standard errors.
  - We can control for other variables which may reduce the residual variance (lead to smaller standard errors).
  - It is easy to include multiple periods.
  - We can study treatments with different treatment intensity. (e.g. varying increases in the minimum wage for different states).

Regression DD

- The typical regression model that we estimate is:

$$Outcome_{it} = \beta_1 + \beta_2 Treat_i + \beta_3 Post_t + \beta_4 (Treat \times Post)_{it} + \epsilon$$

- Treatment $=$ a dummy if the observation is in the treatment group
- Post $=$ post treatment dummy

Regression DD-Card & Krueger

- In the Card & Krueger case the equivalent regression model would be:

$$Y_{ist} = \alpha + \gamma NJ_s + \lambda d_t + \delta(NJ_s \times d_t) + \epsilon_{ist}$$

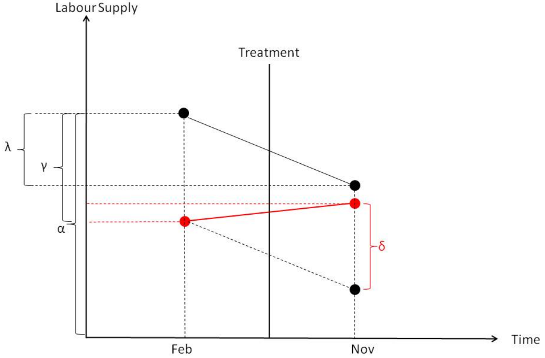
  - NJ is a dummy which is equal to 1 if the observation from NJ.
  - d is a dummy which is equal to 1 if the observation is from November (post).

- This equation takes the following values:
  - PA Pre: α PA Post: $\alpha + \lambda$
  - NJ Pre: $\alpha + \gamma$ NJ Post:$\alpha + \gamma + \lambda + \delta$

- Diff in Diff estimate: (NJ Post - NJ Pre) - (PA Post - PA Pre) $= \delta$

Graph DD

$$Y_{ist} = \alpha + \gamma NJ_s + \lambda d_t + \delta(NJ_s \times d_t) + \epsilon_{ist}$$



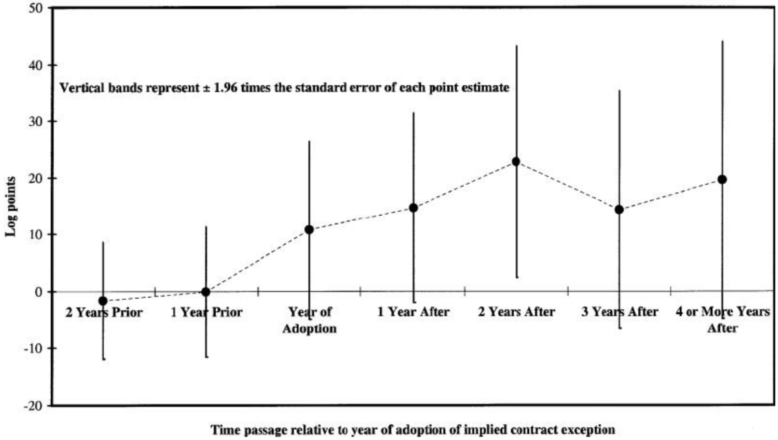
Dynamic Analysis

- Having repeated years given you opportunity to test the causality of your estimation

$$Y_{ist} = \gamma_s + \lambda_t + \sum_{\tau=0}^{m} \beta_{-\tau} D_{s,t-\tau} + \sum_{\tau=1}^{q} \beta_{+\tau} D_{s,t+\tau} + X_{ist}\delta + \epsilon_{ist}$$

- The m lags are called post treatment effects. q leads are anticipatory effects
- We may effect that the effects should grow or fade as time passes.

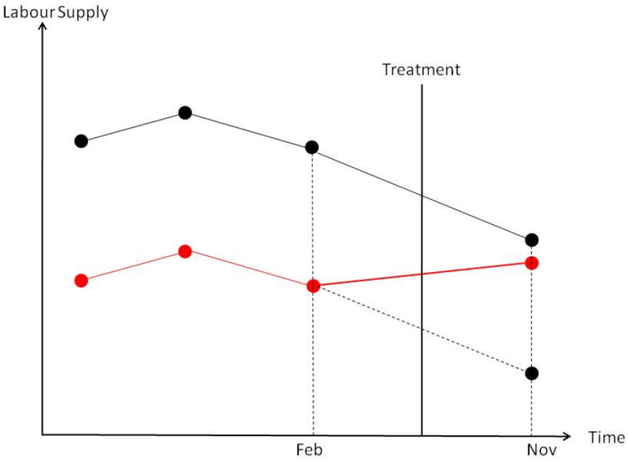
Dynamic Analysis



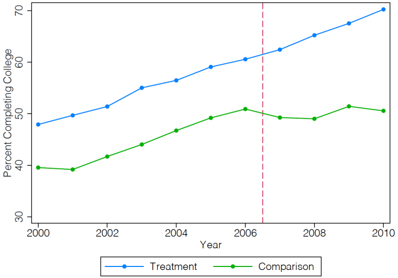
Key assumption of any DD strategy: Common trends/Parallel Trend

- The key assumption for any DD strategy is that the outcome in treatment and control group would follow the same time trend in the absence of the treatment.
- This does not mean that they have to have the same mean of the outcome!
- Common trend assumption is difficult to verify but one often uses pre-treatment data to show that the trends are the same.
- Even if pre-trends are the same one still has to worry about other policies changing at the same time.

Common Trends/Parallel Trend



Common Trends/Parallel Trend



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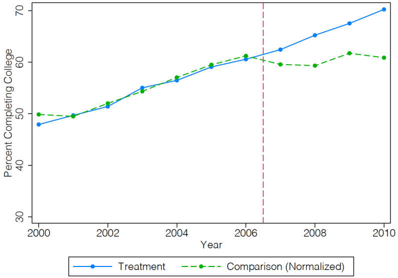
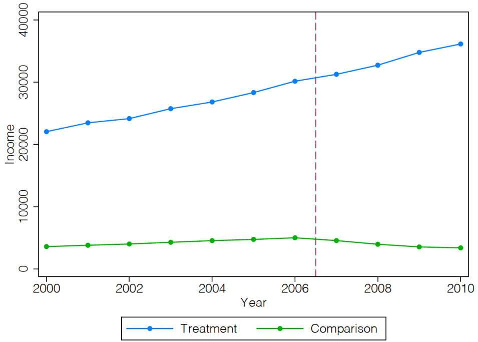
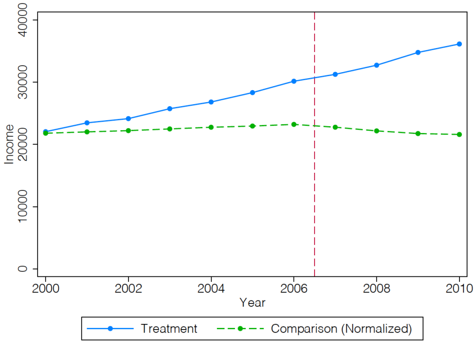


Sometimes the common trend assumption is clearly visible

Common Trends/Parallel Trend

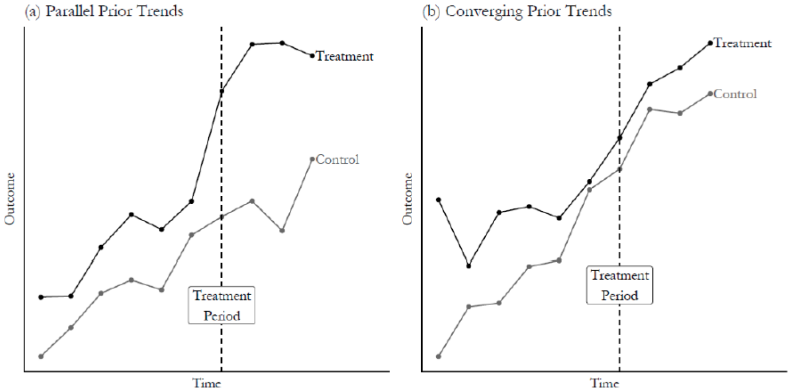


$\Rightarrow$



Sometimes the common trend assumption is clearly violated

Common Trends/Parallel Trend



Defending common trend assumption

Three approaches

- A compelling graph
- A falsification test, or analogously, a direct test in panel data
- Controlling for time trends directly

None of these approaches are possible with two periods of data

Other types of DD

The event study framework includes dummies for each post-treatment period:

$$Y_{i,t} = \alpha + \eta_i + \upsilon_t + \gamma_1 D1_{i,t} + \gamma_2 D2_{i,t} + \gamma_3 D3_{it} + ... + \epsilon_{i,t}$$

When treatment intensity is a continuous variable:

$$Y_{i,t} = \alpha + \beta Intensity_i + \xi Post_t + \delta(Intensity_i \times Post_t) + \epsilon_{i,t}$$

DiD in Practice (WB, 2021)

- Recent publication by the WB provides a checklist in doing DD estimation
  - Argue the parallel trend. If two rounds of data are available before the start of the program, test to see if any difference in trends appears between the two groups.
  - Perform robustness analysis using several plausible comparison groups
  - Perform falsification test using (1) fake outcome or (2) fake treatment (both are control groups).

Limitation of DiD

- If the common trend assumption fails, the results are most likely to be biased
- Failing to account for any factors that affect the difference in trends between the two groups will lead to biased estimates.
- Difficult to control all other policies that are happening at the same time

Example: A Natural Experiment in Education

Esther Duflo studies the impacts of a large school construction program in Indonesia

Schooling and Labor Market Consequences of School Construction in Indonesia: Evidence from an Unusual Policy Experiment

Esther Duflo

AMERICAN ECONOMIC REVIEW
VOL. 91, NO. 4, SEPTEMBER 2001
(pp. 795–813)

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Article Information

Abstract

Between 1973 and 1978, the Indonesian government engaged in one of the largest school construction programs on record. Combining differences across regions in the number of schools constructed with differences across cohorts induced by the timing of the program suggests that each primary school constructed per 1,000 children led to an average increase of 0.12 to 0.19 years of education, as well as a 1.5 to 2.7 percent increase in wages. This implies estimates of economic returns to education ranging from 6.8 to 10.6 percent.

Example: A Natural Experiment in Education

The Sekolah Dasar INPRES program (1973-1979)

- Oil crisis creates large windfall for Indonesia
- Suharto uses oil money to fund school construction
- Close to 62,000 schools built by national government
  - Approximately 1 school built per 500 school-age children
- More schools built in areas which started with fewer schools
- Schools intended to promote equality, national identity

Example: A Natural Experiment in Education

- Do children who were born into areas with more newly built INPRES primary schools get more education? Do they earn more as adults?
- Strategy: difference-in-differences estimation
    - Data on children born before and after program (pre vs. post)
        - Children aged 12 and up in 1974 did not benefit from program
        - Children aged 6 and under were young enough to be treated
    - Data on children born in communities where many schools were built (treatment), those where few schools were built (comparison)
        - Partition sample based on residuals from a regression of the number of schools built (per district) on the number of school-aged children
    - Difference-in-differences estimate of program impact compares pre vs. post differences in treatment vs. comparison communities

Example: A Natural Experiment in Education

The simplest DD estimator is:

$$DD = \bar{Y}^{treatment}_{post} - \bar{Y}^{treatment}_{pre} - (\bar{Y}^{comparison}_{post} - \bar{Y}^{comparison}_{pre})$$

Dependent Variable: Years of Schooling

	Many schools Built	Few Schools Built	Difference
Over 11 in 1974	8.02	9.40	-1.38
Under 7 in 1974	8.49	9.76	-1.27
Difference	0.47	0.36	0.12

Diff-in-Diff estimation compares the change in years of schooling (i.e. the pre vs. post estimate)in treatment, control areas

- Program areas increased faster than comparison areas

Example: A Natural Experiment in Education

The simplest DD estimator is:

$$DD = \bar{Y}^{treatment}_{post} - \bar{Y}^{treatment}_{pre} - (\bar{Y}^{comparison}_{post} - \bar{Y}^{comparison}_{pre})$$

Dependent Variable: Log (Wages)

	Many schools Built	Few Schools Built	Difference
Over 11 in 1974	6.87	7.02	-0.15
Under 7 in 1974	6.61	6.73	-0.12
Difference	-0.26	-0.29	0.026

Diff-in-Diff estimation compares the change in years of schooling (i.e. the pre vs. post estimate)in treatment, control areas

- Program had a modest impact on adult wages

Variation in Treatment Timing

Example: counties introduced food stamps at different times

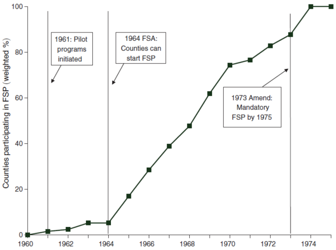


Figure 1. Weighted Percent of Counties with Food Stamp Program, 1960–1975

Source: Authors' tabulations of food stamp administrative data (US Department of Agriculture, various years). Counties are weighted by their 1960 population.

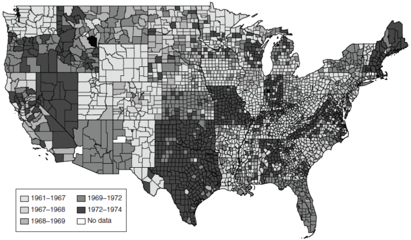


Figure 2. Food Stamp Program Start Date, by County, 1961–1974

Notes: Authors' tabulations of food stamp administrative data (US Department of Agriculture, various years). The shading corresponds to the county FSP start date, where darker shading indicates later county implementation.

Source: Almond, Hoynes, and Schanzenbach (AER, 2016)

Variation in Treatment Timing

Example: States adopted Medicaid at different times

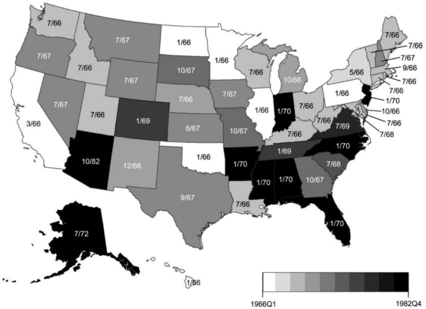


Figure 2.
Medicaid Adoption by Quarter
Notes: Adoption dates come from the Department of Health Education and Welfare (1970) & Social Security Administration (2013). The map is shaded relative to the quarter of adoption and states are labeled with the month and year of adoption.

Source: Boudreaux, Golberstein, and McAlpine (Journal of Health Economics, 2016)

Variation in Treatment Timing

Example: Counties opened community health centers at different times

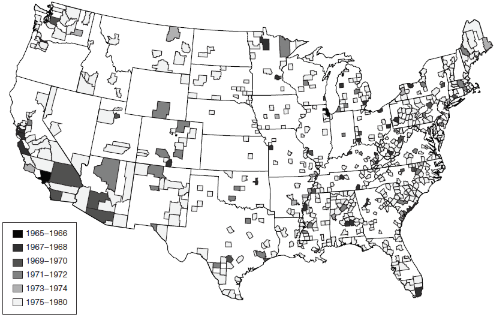


FIGURE 3. ESTABLISHMENT OF COMMUNITY HEALTH CENTERS BY COUNTY OF SERVICE DELIVERY, 1965–1980

Note: Dates are the first year that a CHC was established in the county.
Source: Information on CHCs drawn from NACAP and PHS reports.

Source:Bailey and Goodman-Bacon (AER, 2015)

Fixed Effects-DiD

- If you have longitudinal data, FE-DD helps you to have more unbiased estimate as it helps to purge bias associated with individual's unobserved characteristics

$$Y_{ist} = \alpha_i + \gamma_s + \lambda_t + \beta D_{s,t} + \epsilon_{ist}$$

Fixed Effects Estimates of β^{DD}

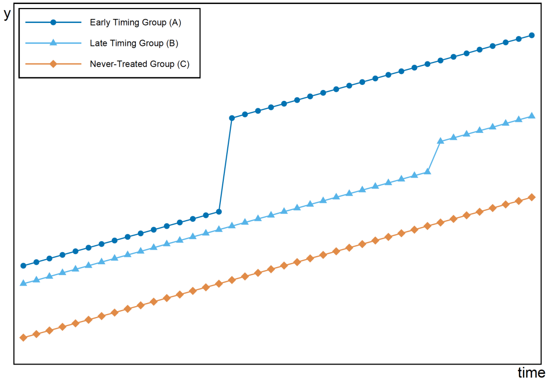
$$Y_{it} = \alpha_i + \gamma_t + \beta^{DD} D_{it} + \epsilon_{it}$$

Where D_{it} is the continuous treatment variable, α_i: individual/state fixed effects and γ_t: time fixed effects

Staggered DD

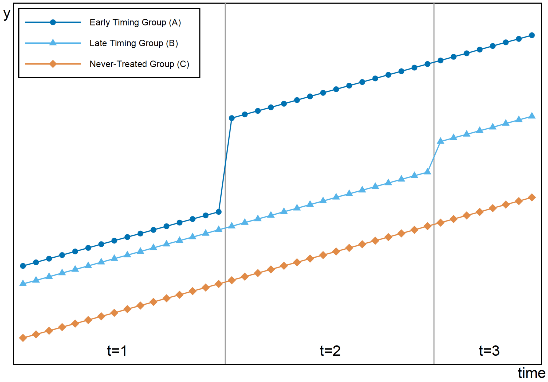
- Some policies are introduced in staggered manner. Leads to variation in timing
  - Roads were built in phases
  - Jurisdictions hand down legal rulings
  - Local governments change policy
- Goodman-Bacon (2018): DID estimate is a weighted average of all possible two-group/two-period DD estimators in the data
- Important analysis when you want to do robustness analysis

Decomposition into Timing Groups



Goodman-Bacon (2019): panel with variation in treatment timing can be decomposed into timing groups reflecting observed onset of treatment

Decomposition into Timing Groups



Example: with three timing groups (one of which is never treated), we can construct three timing windows (pre, middle, post or $t = 1, 2, 3$)